

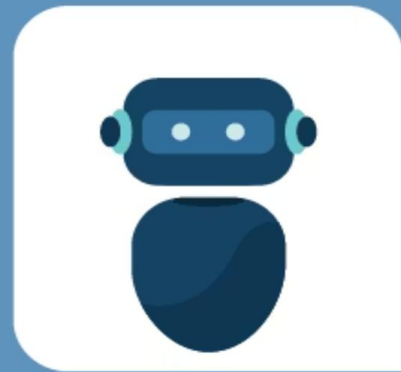
RagStack - Unified Multi-File Vector Knowledge System

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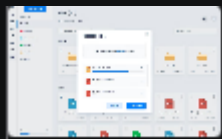


Problem Statement

- Knowledge scattered across heterogeneous formats (PDF, DOCX, images, CSV, JSON).
- AI models cannot directly consume raw files—format & noise barriers.
- No lightweight, universal pipeline to produce portable RAG-ready knowledge bases.
- Need: a robust ingestion system that normalizes multi-format sources into AI-ready vectors.



Solution (High-level)



Multi-file Upload

Unified uploader for heterogeneous inputs and batch ingestion.



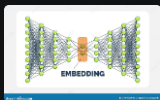
Universal Processor & OCR

Format-specific parsers + OCR pipeline for scanned images.



Automatic Chunking

Semantic-aware splitting with overlap to preserve context.



Embedding Generation

SentenceTransformer embeddings for dense retrieval.



Portable SQLite RAG DB

Single-file .db export for offline sharing and reproducibility.



Retriever & Validation

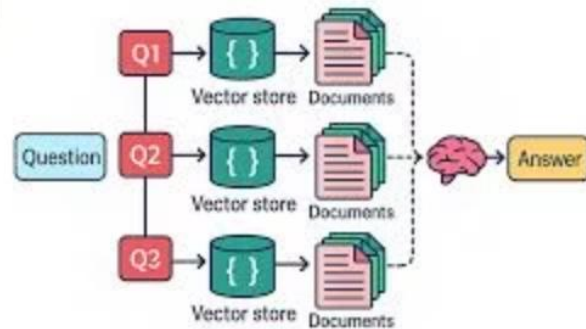
Cosine-similarity retriever with provenance checks for reliable answers.

System Architecture



This diagram shows the production flow from ingestion to retrieval-ready knowledge base and downstream LLM consumption.

Skill
Curl





Key Features & Advantages

- Multi-format support (PDF, DOCX, images, CSV, JSON)
- AI-ready vectorization with SentenceTransformer
- Portable single-file SQLite .db export
- Offline-first: works without cloud dependencies
- Scalable ingestion: batch + streaming-friendly
- Interoperable: integrates with any AI agent or LLM

Applications & Future Scope

Applications

- Academic knowledge systems & literature indexing
- Enterprise document AI and compliance search
- Research repositories and dataset catalogs

Future Scope

- Source-aware retrieval & stronger provenance
- Hybrid search: sparse + dense fusion
- API deployment & large-scale vector engine integration

RAGForge enables reproducible, portable RAG pipelines—designed for hackathons, academic demos, and production prototyping.

